

JUnit Basic Testing

Exercise 1: Setting Up JUnit

Problem

Set up JUnit in a Java project to enable unit testing.

Setup

Create a Java project, add JUnit 4.13.2 dependency, and create a test class.

Java Implementation

```
import org.junit.Test;
import static org.junit.Assert.*;
```

```
public class SampleTest {
    @Test
    public void testSetup() {
        assertTrue(true);
    }
}
```

Output

Program executes successfully and all JUnit test cases pass.

Exercise 2: Writing Basic JUnit Tests

Problem

Write unit tests for a simple Calculator class.

Setup

Create Calculator.java and CalculatorTest.java.

Java Implementation

```
public class Calculator{
    public int add(int a,int b){ return a+b; }
    public int sub(int a,int b){ return a-b; }
}
```

```
import org.junit.Test;
import static org.junit.Assert.*;
```

```
public class CalculatorTest{
    Calculator c=new Calculator();
```

```
@Test
public void testAdd(){
    assertEquals(10,c.add(7,3));
}
```

```
@Test
public void testSub(){
    assertEquals(4,c.sub(7,3));
}
}
```

Output

Program executes successfully and all JUnit test cases pass.

Exercise 3: Assertions in JUnit

Problem

Use different JUnit assertions to validate results.

Setup

Create a test class using common assertion methods.

Java Implementation

```
import org.junit.Test;
import static org.junit.Assert.*;
```

```
public class AssertionsTest{
    @Test
    public void testAssertions(){
        assertEquals(5,2+3);
        assertTrue(5>3);
        assertFalse(5<3);
        assertNull(null);
        assertNotNull(new Object());
    }
}
```

Output

Program executes successfully and all JUnit test cases pass.

Exercise 4: AAA Pattern, Setup and Teardown

Problem

Organize tests using Arrange-Act-Assert with @Before and @After.

Setup

Use setup and teardown methods to initialize and clean resources.

Java Implementation

```
import org.junit.*;
import static org.junit.Assert.*;

public class CalculatorAAATest{

    Calculator c;

    @Before
    public void setUp(){
        c=new Calculator();
    }

    @After
    public void tearDown(){
        c=null;
    }

    @Test
    public void testAdd(){
        // Arrange
        int a=10,b=5;

        // Act
        int result=c.add(a,b);

        // Assert
        assertEquals(15,result);
    }
}
```

Output

Program executes successfully and all JUnit test cases pass.